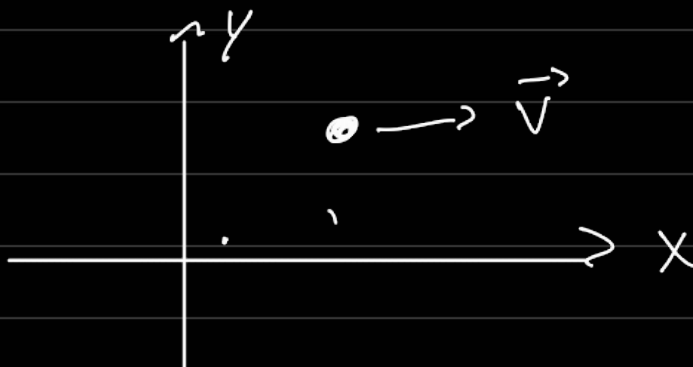
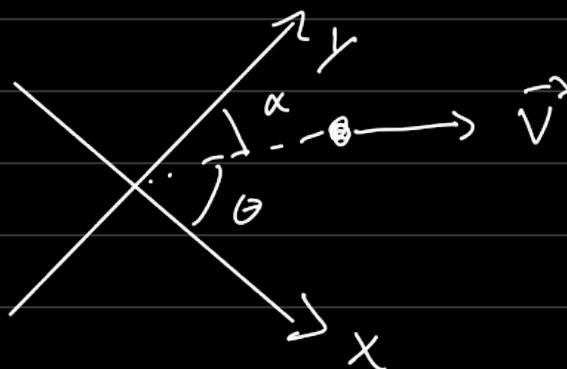


* physics $\longleftrightarrow$ Maths
 "real world" "language"

* I choose:

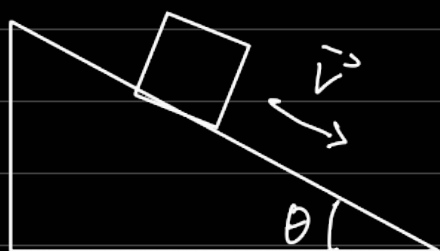


* Someone else:



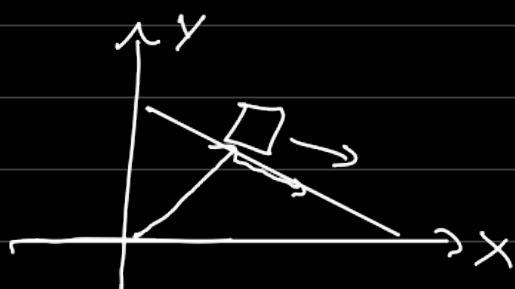
* In general: choose a reference frame to suit the problem

ex:



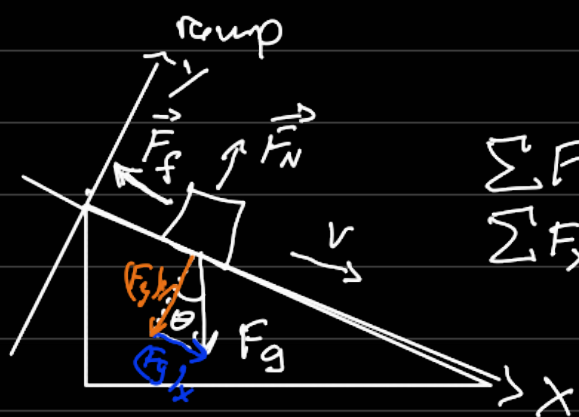
Q: Forces, vel.

(i) we could use



• But: too complicated

(ii) Instead: set up coord system on the



$$\sum F_x = -F_f + F_g \sin(\theta) = ma$$

$$\sum F_y = -F_g \cos(\theta) + F_N = 0$$

$$F_N = F_g \cos(\theta)$$

$$F_f = F_g \sin(\theta) - ma$$

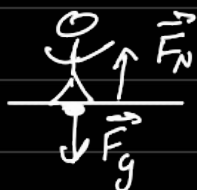
• Newton's laws: $\sum \vec{F} = m\vec{a}$

• Frictional forces:

→ resistive force: opposes "motion"

Normal force:

• Newton: every action $\leftrightarrow$ equal and opposite

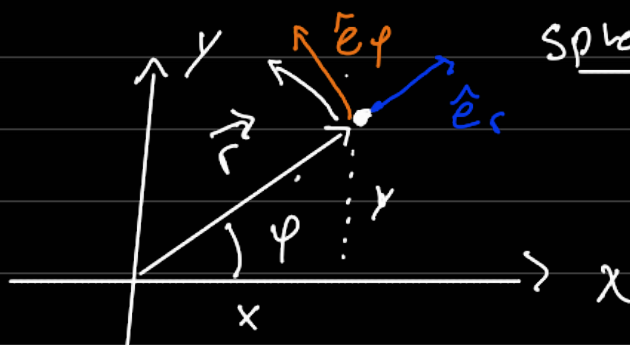


* Vector valued functions

$$\begin{aligned} \vec{r}(t) &= r_x(t) \hat{e}_x + r_y(t) \hat{e}_y + r_z(t) \hat{e}_z \\ &= x(t) \hat{e}_x + y(t) \hat{e}_y + z(t) \hat{e}_z \end{aligned}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \dot{x}(t) \hat{e}_x + \dot{y}(t) \hat{e}_y + \dot{z}(t) \hat{e}_z$$

Circular motion:



(3D) Spherical coordinates / (2D) polar coords

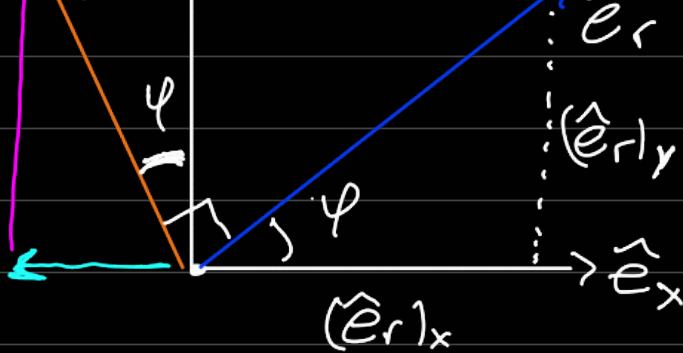
$$x = r \cos(\varphi)$$

$$y = r \sin(\varphi)$$

2D: $\vec{r} = x \hat{e}_x + y \hat{e}_y = r \cos(\varphi) \hat{e}_x + r \sin(\varphi) \hat{e}_y$

$$\dot{x}^2 = \dot{x} \cdot \dot{x} = \frac{dx}{dt} \frac{dx}{dt} = \left(\frac{dx}{dt} \right)^2 \neq \frac{d^2x}{dt^2} = \ddot{x}$$





$$\hat{e}_r = \cos(\varphi) \hat{e}_x + \sin(\varphi) \hat{e}_y$$

$$\hat{e}_\varphi = -\sin(\varphi) \hat{e}_x + \cos(\varphi) \hat{e}_y$$

$$\begin{aligned} \hookrightarrow (\hat{e}_\varphi)_x &= -\sin(\varphi) \\ (\hat{e}_\varphi)_y &= \cos(\varphi) \end{aligned}$$

$$\begin{aligned} \vec{r} &= r \cos(\varphi) \hat{e}_x + r \sin(\varphi) \hat{e}_y \\ &= r (\cos(\varphi) \hat{e}_x + \sin(\varphi) \hat{e}_y) \end{aligned}$$

$$\vec{r} = r \hat{e}_r$$

* Side note: $\hat{e}_r = \hat{r}$ or $\hat{e}_\varphi = \hat{\varphi}$

• Want to know $\frac{d\hat{e}_r}{d\varphi}$ and $\frac{d\hat{e}_\varphi}{d\varphi}$:

$\rightarrow$ Basis vectors change $\hat{e}_r$ & $\hat{e}_\varphi$:

$$\text{(i) look at: } \frac{d\hat{e}_r}{d\varphi} = \frac{d}{d\varphi} (\cos(\varphi) \hat{e}_x + \sin(\varphi) \hat{e}_y)$$

$$\frac{d\hat{e}_r}{d\varphi} = -\sin(\varphi) \hat{e}_x + \cos(\varphi) \hat{e}_y = \hat{e}_\varphi$$

$$\frac{d\hat{e}_r}{d\varphi} = \hat{e}_\varphi \quad *$$

$$\text{(ii) look at } \frac{d\hat{e}_\varphi}{d\varphi} = \frac{d}{d\varphi} (-\sin(\varphi) \hat{e}_x + \cos(\varphi) \hat{e}_y)$$

$$\frac{d\hat{e}_\varphi}{d\varphi} = -\cos(\varphi) \hat{e}_x - \sin(\varphi) \hat{e}_y = -(\cos(\varphi) \hat{e}_x + \sin(\varphi) \hat{e}_y)$$

$$\frac{d\hat{e}_\varphi}{d\varphi} = -\hat{e}_r$$

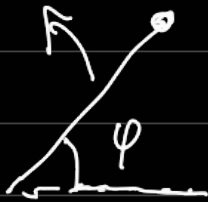
$\hat{e}_r$

* Put this to use: $\vec{r} = r \hat{e}_r$ (Position)

→ Velocity: $\frac{d\vec{r}}{dt} = \dot{\vec{r}} = \vec{v}$ *

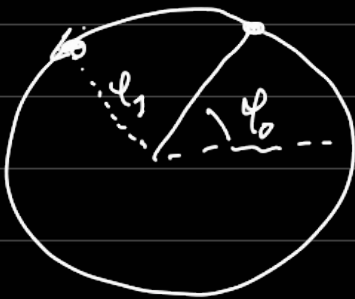
* Circular motion: angular velocity: $\frac{d\varphi}{dt} = \omega$

^{1D}
(linear vel. $\frac{dx}{dt} = v$)



$\frac{\text{miles}}{\text{hour}} \leftarrow \frac{\text{distance}}{\text{time}} \sim \frac{x}{t} = v$

ex: 80 mph → 1 hour they travel 80 miles



$\frac{d\varphi}{dt} \leftarrow \begin{matrix} \text{"distance"} \\ \text{"time"} \end{matrix} \rightarrow \omega$

$$\begin{aligned} \frac{d\vec{r}}{dt} = \vec{v}(t) &= \frac{d}{dt}(r \hat{e}_r) = \frac{dr}{dt} \hat{e}_r + r \frac{d\hat{e}_r}{dt} \\ &= \dot{r} \hat{e}_r + r \frac{d\hat{e}_r}{dt} \end{aligned}$$

$\frac{d\hat{e}_r}{dt}$, chain rule: $\hat{e}_r(\varphi)$

φ : changes over time

* $\frac{d\hat{e}_r}{dt} = \frac{d\varphi}{dt} \frac{d\hat{e}_r(\varphi)}{d\varphi} = \dot{\varphi} \hat{e}_\varphi$ *

plug in

$$d\vec{r} = \dot{r} \hat{e}_r + r \dot{\varphi} \hat{e}_\varphi = \vec{v} \quad (\text{velocity})$$

$$\frac{d}{dt} = \dot{r} \hat{e}_r + r (\dot{\varphi} \hat{e}_\varphi) = v \quad (\text{velocity})$$

* Finally: acceleration: $\frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

$$\begin{aligned} \frac{d^2\vec{r}}{dt^2} &= \frac{d}{dt} (\dot{r} \hat{e}_r + r \dot{\varphi} \hat{e}_\varphi) \\ &= \left[\frac{d\dot{r}}{dt} \hat{e}_r + \dot{r} \frac{d\hat{e}_r}{dt} \right] + \left[\frac{dr}{dt} \dot{\varphi} \hat{e}_\varphi + r \frac{d\dot{\varphi}}{dt} \hat{e}_\varphi + r \dot{\varphi} \frac{d\hat{e}_\varphi}{dt} \right] \\ &= [\ddot{r} \hat{e}_r + \dot{r} \dot{\varphi} \hat{e}_\varphi] + [\dot{r} \dot{\varphi} \hat{e}_\varphi + r \ddot{\varphi} \hat{e}_\varphi + r \dot{\varphi} \frac{d\hat{e}_\varphi}{dt}] \end{aligned}$$

$$\bullet \frac{d\hat{e}_\varphi}{dt} = \frac{d\varphi}{dt} \underbrace{\frac{d\hat{e}_\varphi}{d\varphi}}_{-\hat{e}_r} = -\dot{\varphi} \hat{e}_r$$

Plug in:

$$= [\ddot{r} \hat{e}_r + \dot{r} \dot{\varphi} \hat{e}_\varphi] + [\dot{r} \dot{\varphi} \hat{e}_\varphi + r \ddot{\varphi} \hat{e}_\varphi - r \dot{\varphi} \dot{\varphi} \hat{e}_r]$$

now collect basis vectors

$$= (\ddot{r} - r(\dot{\varphi})^2) \hat{e}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi}) \hat{e}_\varphi \quad (\text{acceleration})$$

$$\vec{a} = (\ddot{r} - r(\dot{\varphi})^2) \hat{e}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi}) \hat{e}_\varphi$$

PS3: Q3 | $r(\varphi) = a e^{k\varphi}$

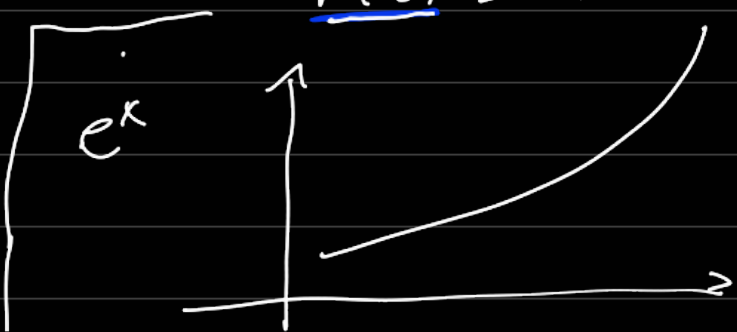
• Const ang vel. $\frac{d\varphi}{dt} = \omega$

integrate: $\varphi = \omega t$

plug in:

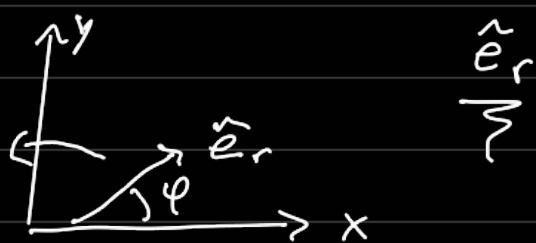
$$r(t) = a e^{k\omega t}$$

$$\vec{r} = r \hat{e}_r = x \hat{e}_x + y \hat{e}_y$$



$$\varphi = \omega t$$

$$a e^{k\omega t} \hat{e}_r = r(\cos(\varphi) \hat{e}_x + \sin(\varphi) \hat{e}_y)$$



$$\vec{r} = a e^{k\omega t} \hat{e}_r \quad | \quad r = a e^{k\omega t}$$

$$\vec{F} = m \vec{a}$$

$$\vec{a} = (\ddot{r} - r\dot{\varphi}^2) \hat{e}_r + (r\ddot{\varphi} + 2\dot{r}\dot{\varphi}) \hat{e}_\varphi$$

$$r = a e^{k\omega t}$$

$$\dot{r} = a\omega k e^{k\omega t} = \omega k r$$

$$\ddot{r} = (\omega k)^2 a e^{k\omega t} = (\omega k)^2 r$$

$$\varphi = \omega t$$

$$\dot{\varphi} = \omega$$

$$\ddot{\varphi} = 0$$

$$\begin{aligned} \vec{a} &= ((\omega k)^2 r - r(\omega)^2) \hat{e}_r + (\cancel{r\ddot{\varphi}}^0 + 2\omega k r \omega) \hat{e}_\varphi \\ &= r\omega^2(k^2 - 1) \hat{e}_r + 2\omega^2 k r \hat{e}_\varphi \end{aligned}$$

$$\vec{F} = m (r\omega^2(k^2 - 1) \hat{e}_r + 2\omega^2 k r \hat{e}_\varphi)$$

